D1. Causative-inchoative alternation

Model and Date

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Engineered Prompt:

You are a linguist, and you are doing token-level semantic role labelling. In this task, ONLY the chunk heads gets labelled. You are interested in <ARG1> role invariance in verb causative-inchoative alternation. Here is an example:

- *I broke <X> the window <ARG1> .*
- *The window <ARG1> broke <V> .*

As you can see, the semantic role of `window' is unchanged regardless of alternation.

You need to generate 50 more sentence pairs like the example demonstrated. Requirements:

1) *the predicate should allow causative-inchoative alternation, and <ARG1>.*
2) *The generated use cases should be both grammatically and logically correct.*
3) *There is always a space between punctuation and a word.*

*In total, you generate *100 instances*. The final output format is in the. tsv format with **four** columns:*

1) *use cases,*
2) *predicate and target in focus, separated by a semicolon, i.e., broke; window*
3) *predicate label per token (incl. punctuation) separated by commas, using '_' (not predicate) or 'X' (predicate)*
4) *Argument labels (separated by a comma), where only ARG1 is labelled, and the rest are O.*

Make sure the length of the predicate and argument labels match the number of tokens per use case.

.Example filtering/modification criteria:

1) The predicate allows alternation, and <ARG1>
2) There are no mistakes (e.g. grammatical, spellings), logic issues, and formatting problems in the updated use cases.
3) The length of the annotation matches the nr of tokens.
4) Predicates are annotated correctly.
5) <ARG1> are annotated correctly following PropBank conventions.

Gemini Response:

> ***Defining the Core Focus***
> *I now understand the core goal: generating causative-inchoative sentence pairs with an invariant theme/patient.*
> ***Refining the Output Format***
> *I've established the output format, token labeling, punctuation spacing, and crucial generation constraints.*
> ***Compiling Verbs for Dataset***
> *I've curated a list of alternating causative-inchoative verbs, ready for sentence pair generation.*
> ***Validating Verbs List***
> *I've tested and confirmed the causative/inchoative verbs, ensuring the patient is consistently ARG1.*
> ***Expanding the Verb List***
> *I've expanded the curated list, carefully checking all 50 entries for multi-token and structural issues.*
> ***Generating Sentence Pairs***
> *I'm now building the final rows for sentence pairs, including varied subjects for causatives.*
> ***Creating Sentence Pairs***
> *I am now generating sentence pairs with coreferential arguments, ensuring correct semantic roles.*
> ***Generating More Sentence Pairs***
> *I'm now generating additional causative-inchoative sentence pairs, expanding the dataset steadily.*
> ***Creating More Sentence Pairs***
> *I'm now generating additional causative-inchoative sentence pairs, expanding the dataset steadily.*
> *I have generated 10 further sentence pairs, carefully checking for consistency of role labeling.*
> ***Adding More Sentence Pairs***
> *I've generated 13 more high-quality causative-inchoative sentence pairs for the dataset.*
> ***Formatting the TSV Output***
> *I've crafted and meticulously verified the TSV output structure, and am ready to generate the complete dataset.*

This is a great task. Focusing exclusively on chunk heads for semantic role labeling while ensuring strict token alignment can be tedious, but causative-inchoative alternations are perfect for demonstrating <ARG1> role invariance.

Below is the TSV-formatted dataset with 50 pairs (100 instances) that meet all your requirements. The token counts perfectly match the label sequences, and every punctuation mark is spaced as an independent token.

Use Case	Predicate; Target	Predicate Labels	Argument Labels
I melted the ice .	melted; ice	_, X, _, _, _	O, O, O, ARG1, O
The ice melted .	melted; ice	_, _, X, _	O, ARG1, O, O
You opened the door .	opened; door	_, X, _, _, _	O, O, O, ARG1, O
The door opened .	opened; door	_, _, X, _	O, ARG1, O, O
He closed the gate .	closed; gate	_, X, _, _, _	O, O, O, ARG1, O
The gate closed .	closed; gate	_, _, X, _	O, ARG1, O, O

She sank the ship . sank; ship _, X, _, _, _ O, O, O, ARG1, O

The ship sank . sank; ship _, _, X, _ O, ARG1, O, O

We shattered the glass . shattered; glass _, X, _, _, _ O, O, O, ARG1, O

The glass shattered . shattered; glass _, _, X, _ O, ARG1, O, O

They boiled the water . boiled; water _, X, _, _, _ O, O, O, ARG1, O

The water boiled . boiled; water _, _, X, _ O, ARG1, O, O

John bounced the ball . bounced; ball _, X, _, _, _ O, O, O, ARG1, O

The ball bounced . bounced; ball _, _, X, _ O, ARG1, O, O

Mary rolled the stone . rolled; stone _, X, _, _, _ O, O, O, ARG1, O

The stone rolled . rolled; stone _, _, X, _ O, ARG1, O, O

Bob dropped the cup . dropped; cup _, X, _, _, _ O, O, O, ARG1, O

The cup dropped . dropped; cup _, _, X, _ O, ARG1, O, O

Alice started the engine . started; engine _, X, _, _, _ O, O, O, ARG1, O

The engine started . started; engine _, _, X, _ O, ARG1, O, O

Tom stopped the car . stopped; car _, X, _, _, _ O, O, O, ARG1, O

The car stopped . stopped; car _, _, X, _ O, ARG1, O, O

Jane spun the wheel . spun; wheel _, X, _, _, _ O, O, O, ARG1, O

The wheel spun . spun; wheel _, _, X, _ O, ARG1, O, O

I bent the wire . bent; wire _, X, _, _, _ O, O, O, ARG1, O

The wire bent . bent; wire _, _, X, _ O, ARG1, O, O

You froze the meat . froze; meat _, X, _, _, _ O, O, O, ARG1, O

The meat froze . froze; meat _, _, X, _ O, ARG1, O, O

He cooked the rice . cooked; rice _, X, _, _, _ O, O, O, ARG1, O

The rice cooked . cooked; rice _, _, X, _ O, ARG1, O, O

She dissolved the sugar . dissolved; sugar _, X, _, _, _ O, O, O, ARG1, O

The sugar dissolved . dissolved; sugar _, _, X, _ O, ARG1, O, O

We dried the shirt .dried; shirt _, X, _, _, _ O, O, O, ARG1, O

The shirt dried . dried; shirt _, _, X, _ O, ARG1, O, O

They expanded the balloon . expanded; balloon _, X, _, _, _ O, O, O, ARG1, O

The balloon expanded .expanded; balloon _, _, X, _ O, ARG1, O, O

John gathered the crowd . gathered; crowd _, X, _, _, _ O, O, O, ARG1, O

The crowd gathered . gathered; crowd _, _, X, _ O, ARG1, O, O

Mary healed the wound . healed; wound _, X, _, _, _ O, O, O, ARG1, O

The wound healed . healed; wound _, _, X, _ O, ARG1, O, O

Bob increased the price . increased; price _, X, _, _, _ O, O, O, ARG1, O

The price increased . increased; price _, _, X, _ O, ARG1, O, O

Alice decreased the value . decreased; value _, X, _, _, _ O, O, O, ARG1, O

The value decreased . decreased; value _, _, X, _ O, ARG1, O, O

Tom awakened the boy . awakened; boy _, X, _, _, _ O, O, O, ARG1, O

The boy awakened . awakened; boy _, _, X, _ O, ARG1, O, O

Jane burned the wood .burned; wood _, X, _, _, _ O, O, O, ARG1, O

The wood burned .burned; wood _, _, X, _ O, ARG1, O, O

I burst the pipe . burst; pipe _, X, _, _, _ O, O, O, ARG1, O

The pipe burst . burst; pipe _, _, X, _ O, ARG1, O, O

You cleared the screen . cleared; screen _, X, _, _, _ O, O, O, ARG1, O

The screen cleared . cleared; screen _, _, X, _ O, ARG1, O, O

He connected the call . connected; call _, X, _, _, _ O, O, O, ARG1, O

The call connected . connected; call _, _, X, _ O, ARG1, O, O

She crashed the system . crashed; system _, X, _, _, _ O, O, O, ARG1, O

The system crashed . crashed; system _, _, X, _ O, ARG1, O, O

We darkened the room . darkened; room _, X, _, _, _ O, O, O, ARG1, O

The room darkened . darkened; room _, _, X, _ O, ARG1, O, O

They floated the raft . floated; raft _, X, _, _, _ O, O, O, ARG1, O

The raft floated . floated; raft _, _, X, _ O, ARG1, O, O

John flew the kite . flew; kite _, X, _, _, _ O, O, O, ARG1, O

The kite flew . flew; kite _, _, X, _ O, ARG1, O, O

Mary folded the map . folded; map _, X, _, _, _ O, O, O, ARG1, O

The map folded . folded; map _, _, X, _ O, ARG1, O, O

Bob grew the tree . grew; tree _, X, _, _, _ O, O, O, ARG1, O

The tree grew . grew; tree _, _, X, _ O, ARG1, O, O

Alice thawed the chicken . thawed; chicken _, X, _, _, _ O, O, O, ARG1, O

The chicken thawed . thawed; chicken _, _, X, _ O, ARG1, O, O

Tom improved the score . improved; score _, X, _, _, _ O, O, O, ARG1, O

The score improved . improved; score _, _, X, _ O, ARG1, O, O

Jane moved the box . moved; box _, X, _, _, _ O, O, O, ARG1, O

The box moved . moved; box _, _, X, _ O, ARG1, O, O

I poured the juice . poured; juice _, X, _, _, _ O, O, O, ARG1, O

The juice poured . poured; juice _, _, X, _ O, ARG1, O, O

You rang the bell . rang; bell _, X, _, _, _ O, O, O, ARG1, O

The bell rang . rang; bell _, _, X, _ O, ARG1, O, O

He rocked the boat . rocked; boat _, X, _, _, _ O, O, O, ARG1, O

The boat rocked . rocked; boat _, _, X, _ O, ARG1, O, O

She shut the blind . shut; blind _, X, _, _, _ O, O, O, ARG1, O

The blind shut . shut; blind _, _, X, _ O, ARG1, O, O

We snapped the twig . snapped; twig _, X, _, _, _ O, O, O, ARG1, O

The twig snapped . snapped; twig _, _, X, _ O, ARG1, O, O

They spilled the milk . spilled; milk _, X, _, _, _ O, O, O, ARG1, O

The milk spilled . spilled; milk _, _, X, _ O, ARG1, O, O

John split the log . split; log _, X, _, _, _ O, O, O, ARG1, O

The log split . split; log _, _, X, _ O, ARG1, O, O

Mary spread the jam . spread; jam _, X, _, _, _ O, O, O, ARG1, O

The jam spread . spread; jam _, _, X, _ O, ARG1, O, O

Bob stretched the band . stretched; band _, X, _, _, _ O, O, O, ARG1, O

The band stretched . stretched; band _, _, X, _ O, ARG1, O, O

Alice swung the bat . swung; bat _, X, _, _, _ O, O, O, ARG1, O

The bat swung . swung; bat _, _, X, _ O, ARG1, O, O

Tom tore the cloth . tore; cloth _, X, _, _, _ O, O, O, ARG1, O

The cloth tore . tore; cloth _, _, X, _ O, ARG1, O, O

Jane turned the page . turned; page _, X, _, _, _ O, O, O, ARG1, O

The page turned . turned; page _, _, X, _ O, ARG1, O, O

I woke the baby . woke; baby _, X, _, _, _ O, O, O, ARG1, O

The baby woke . woke; baby _, _, X, _ O, ARG1, O, O

You emptied the cup . emptied; cup _, X, _, _, _ O, O, O, ARG1, O

The cup emptied . emptied; cup _, _, X, _ O, ARG1, O, O